

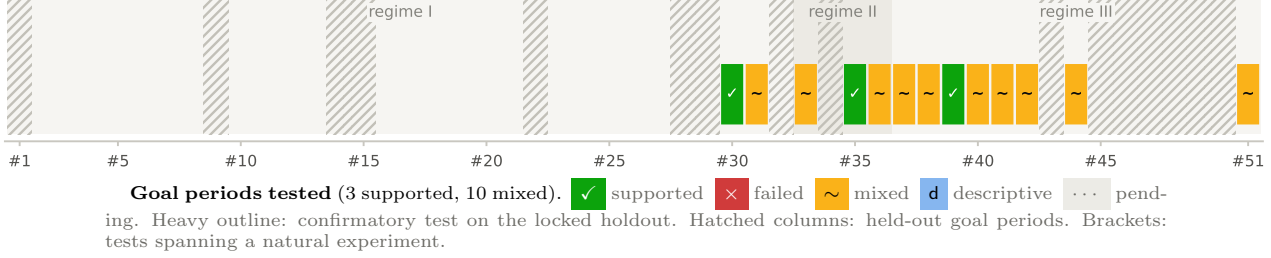
H93: Project choice as a Brock–Durlauf equilibrium

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 1/5 Usefulness 1.5/5

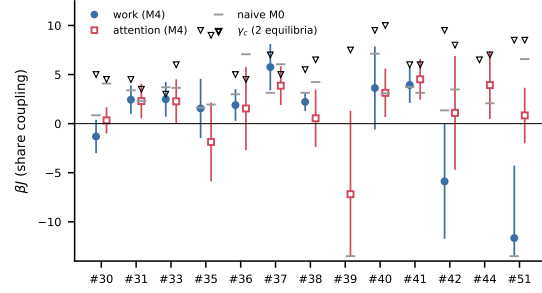
Bottom line. Project choice carries a share coefficient of +2 to +6 in shared weeks and −6 to −12 in own-role weeks, but repo bursts mimic it; at the fitted fields no period is bistable and no same-field replica splits.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

H93 fits project choice as a Brock–Durlauf (BD) logit with a share coupling βJ , per period, on DQ4 work arrivals and attention labels in 13 periods (#30–#51). Multiple equilibria were predicted to split same-type weeks. The estimator M4 controls kickoff naming, habit and repo size; it passes the synthetic false-positive rule on all 13 schedules. $\beta J > 0$ (CI > 0) in 5 of 10 identified work periods, +1.9 to +5.8. It is negative in the own-role weeks (#51 – 11.7 [−19.0, −4.4]). A post hoc synthetic shows that fast repo bursts without coupling give the same values. Every period lies below its multiplicity boundary ($\beta J / \gamma_c \leq 0.86$). No same-field replica lands in a second state. The holdout is not run.



Observables. Share coupling βJ per period (95% CI; DL pool over testable units). Filled circles: work arrivals; open squares: attention switches. Gray dashes: the naive share-only estimate M0. Triangles: γ_c , where the period's fitted fields would first support a second equilibrium. Every point estimate lies below its triangle. The share coefficient is positive in shared-goal weeks and negative in the own-role weeks #42 and #51.

PHYSICAL MODEL

Model 10 (mean-field Potts) in its econ form, with model 05's repo fitness as the confound. An agent's state is its current repo or attention project. A choice event is a switch or a resumption. s_j is the share of the other active agents on option j . The fields are kickoff naming nm_j (H54), habit $prev_{ij}$ (the agent held j before) and log cumulative arrivals c_j (fitness proxy). βJ is the coupling in noise units. γ_c is the coupling at which the fitted fields first support two stable states.

$$P(j | e) \propto \exp(b_n nm_j + b_o prev_{ij} + b_c \ln(1+c_j) + \beta J s_j)$$

$$m_j = \langle \text{softmax}_j(h_i + \beta J m) \rangle_i, \quad \text{stable: } \text{Re } \lambda(\partial F / \partial m) < 1$$

$$\gamma_c = \min\{\beta J : \# \text{stable fixed points} \geq 2\}$$

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Agents do pile onto projects that others already work on: from 0 to a third of the agents multiplies the pick odds by about 2–7 in shared weeks. That term cannot be told apart from a common burst of repo attention, so do not read it as imitation. The useful result is negative: at the fitted fields no week is bistable. An operator need not fear that a free week will lock into a herd by chance. Herding follows the fields: what the kickoff names, which team is assigned, and habit, which is the largest term (odds $\times 12\text{--}400$ of returning to one's own repo). To spread work, change the fields (assign owners, as in #51), not the chat.

RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P1 naive $M0 > 0$	8/10 work periods	supported
P2 fields+habit halve it	0/8 (ratio 0.58–2.8)	failed
P3 M4 CI $\ni 0$ in $\geq 1/2$	3/10; + in 5, – in 2	failed
P4 $P_{\text{multi}} < 0.5$ everywhere	51/54 fits; 3 small units fail	failed (letter)
P5 BD m^* ranks m	Spearman 0.77, 10 periods	supported
P6 work, attention same sign	7/11	failed
P8 BD vs neutral form tie	8/11 within 0.02 nats/event	supported
N G37 rooms, one field	$\Delta m = +0.22$ in band, $p = 0.50$	supported
N G44 free vs assigned	assigned arm +13.9: field absorbed	failed
N G51 12 same-goal units	no herded unit ($m \leq 0.21$); AF –11.7	mixed

GOAL PERIODS IN DETAIL

G30 ✓ supported
 G31 ~ mixed
 G33 ~ mixed
 G35 ✓ supported
 G36 ~ mixed
 G37 ~ mixed
 G38 ~ mixed
 G39 ✓ supported
 G40 ~ mixed
 G41 ~ mixed

+3 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

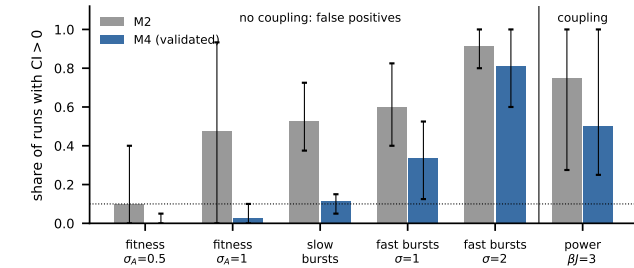
The burst worlds, the read split and the stability fix are post hoc (A2). H54's token rule marks 31–100% of chosen options as kickoff-named, so the kickoff field is measured poorly. An unmodelled field is absorbed into βJ ; #44's assigned team shows it. Work periods have 48–183 choice events outside #51, and M4 power at $\beta J = 3$ reaches 0.8 only in #37, #38 and #51. P_{multi} treats the fitted fields as fixed and the logit's IIA as true. Commits are attempts; attention labels are mentions.

ROUND 2

An in-flight design (hosts the chooser could not yet have seen) or exogenous share shocks (expiries, exits) to identify J ; assignment fields from DQ6 in the model.

- **H93-R1** Separate J from bursts.
- **H93-R2** Put assignment fields in the model.
- **H93-R3** Habit as an agent constant.

H93-brock-durlauf-project-choice · generated 2026-10-07 17:50 by infra/summaries/build_summaries.py · card hypotheses/H93-brock-durlauf-project-choice/README.md · ratings: H93 round-1 agent (suggested) (page 2)



Second figure. Synthetic identification on the real event schedules (40 runs per world; medians and ranges over skeletons). Static repo fitness spread fools the naive M2 but not M4. Fast repo-attention bursts (correlation time 20 arrivals) fool both. Power of M4 at $\beta J = 3$ is 0.25–1.00 by period.